

Week 11 Recitation

TA Suha Kim

ECON2001.01 (Prin. Micro.)

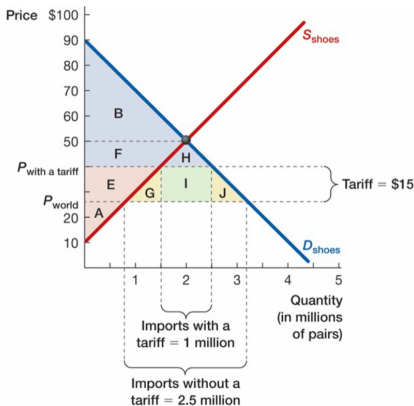
Mar 20, 2025

Chap 8

Protectionism – Tariff

What if there is government intervention for **protectionism**? – Tariff

- Importing country!
- **Tariff**: a tax added onto the price of an imported product
→ So a tariff **increases the market price** of the product



Chap 8

Protectionism – Tariff

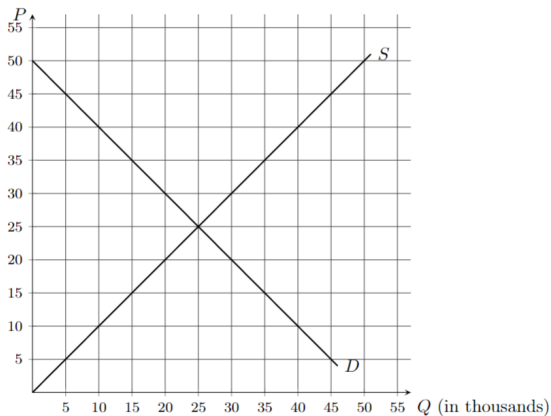
- The location of P_{world} for the country?
- Consumer Surplus
 - Before tariff
 - After tariff
- Producer Surplus
 - Before tariff
 - After tariff
- Government Revenue from tax (tariff)
 - Government Revenue goes to the people in the country through the social welfare policy
 - So it is counted as **Surplus!** ★★★
- Social Surplus / DWL

Chap 8

Example

2. Scenario: The below figure displays Jinjerland's the market supply and demand curves for good Z. Assume the market for good Z is perfectly competitive.

Jinjerland's Market for Good Z



Chap 8

Example – Problem Set

- (a) Assuming a closed economy, what is the equilibrium price and quantity of good Z?
- (b) Suppose the Jinjerlandian government opens its borders to free trade. Will Jinjerland import or export good Z at a world price of \$15?
- (d) Suppose policymakers in Jinjerland levy a \$5 per unit tariff on the import of good Z, causing the price to increase to \$20. Calculate the deadweight loss that arises from the imposition of the tariff.

Externalities

What is Externalities?

Externalities

- An indirect cost or benefit to the third parties who are not directly engaged in the initial transaction (the production and consumption of a good or service)
 - **Negative Externality**
 - Example: Pollution, Noise from the construction site, ...
 - Take tolls on 3rd party
 - **Positive Externality**
 - Example: Education, R&D, Pleasant smell of bread, ...
 - beneficial to 3rd party

Externalities

What is Externalities?

Negative Externality

- Social costs are greater than private costs → Supply side
- Social Costs
= Private Costs + External Costs.
- Marginal Social Costs (MSC) = Marginal Private Costs (MPC) + Marginal External Costs (MEC).
- Not considering MEC
⇒ Overproduction than the socially optimal level.

Positive Externality

- Social benefits are greater than private benefits → Demand side
- Social Benefits
= Private Benefits + External Benefits.
- Marginal Social Benefits (MSB) = Marginal Private Benefits (MPB) + Marginal External Benefits (MEB).
- Not considering MEB
⇒ Underproduction than the socially optimal level.

Externalities

Inefficiency

When externalities exist, social welfare is not maximized at the market equilibrium (**Deadweight loss** occurs).

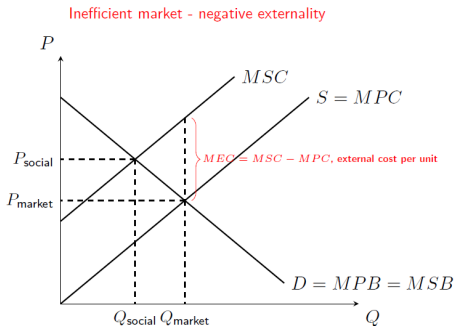
- Social benefits (costs) would be different from the private benefits (costs).
- Individuals only care about their private benefits and costs, not social benefits and costs.
- Market outcome is determined at $P = MPB = MPC$.
- Socially optimal outcome is determined at $P = MSB = MSC$.

Externalities

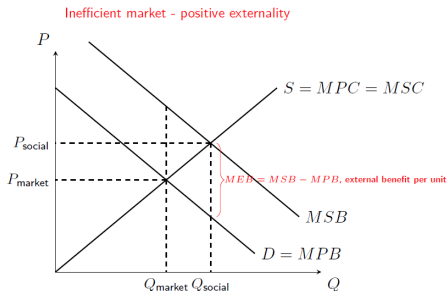
Inefficiency

Let's see how much inefficiency is caused by the externalities

Negative externality



Positive externality



Externalities

Inefficiency

Let's see how much inefficiency is caused by the externalities

- Which curve and How much?
- Socially Optimal
 - Consumer Surplus
 - Producer Surplus
 - Social Surplus
- With Externalities
 - Consumer Surplus
 - Producer Surplus
 - Externalities
 - Social Surplus
- DWL?
- Private Output $\overset{?}{\geq}$ Socially Optimal Output

Externalities

Solutions to Externalities

Private Solutions to Externalities

- For private solutions, individuals should **internalize** the externalities.
 - “I also consider the externalities when making the decisions.”
- **Coase Theorem:** Conditions for private bargaining to be the solution
 - Well-defined property rights
 - Minor transaction costs
 - Small number of individuals engaged in the bargaining process

Externalities

Solutions to Externalities

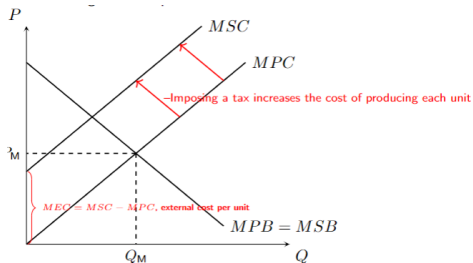
Government Solutions to Externalities

- Command-and-control policies: The government directly regulates the allocation of resources
 - For example, restriction of production level to the socially efficient outcome level in the case of negative externality.
- Market-based policies: The government provides incentives for private agents to internalize the externality.
 - **Pigouvian tax** (Negative externality) / **subsidy** (Positive externality)

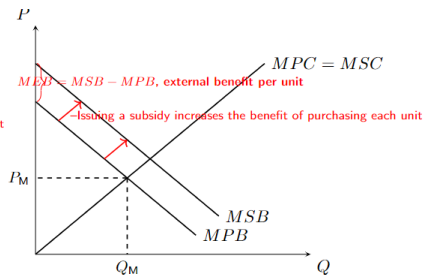
Externalities

Solutions to Externalities

Pigouvian tax



Pigouvian Subsidy



Exercise!

Problem Set

3. (Negative Externality)

Suppose $MPB = 60 - Q$, $MPC = Q$, and there is a negative externality given by $MEC = 20$.

- ① Draw the graph that features MPB , MPC , MEC , MSC , MPC .
- ② Find the private solution (Q_{Priv}) using $MPC=MPB$.
- ③ Find the social solution (Q_{Soc}) using $MSC=MSB$. Note that $MEB=0$ and $MEC>0$ when there is a negative externality.
- ④ Find the deadweight loss, which is the area between MSC and MSB for the quantity discrepancy (Q_P vs. Q_S)
- ⑤ Find the size of the Pigouvian tax to eliminate the negative externality problem.

Public Goods and Common Pool Resource Goods

Classification of goods

- **Rivalry**

- Rival goods: Goods that only one person can consume at a time

- **Excludability**

- Excludable goods: Must be paid for (to exclude not paying people) in order to consume them

Types of goods

	Excludability High	Excludability Low
Rivalry High	Ordinary private goods (e.g., clothes, food, furniture)	Common pool resource goods (e.g., fish, water, natural forests, food at a picnic)
Rivalry Low	Club goods (e.g. cable TV, pay-per-view TV, Wi-Fi, music downloads)	Public goods (e.g., national defense, early warning systems, earth protection programs)

Public Goods and Common Pool Resource Goods

Public Goods

- **Non-rival** and **non-excludable** goods
- Free-rider problem exists:
Positive externalities from other's production $\Rightarrow$
Under-production than socially optimal level.
- Solution: The government provides them.
- Market demand: **Vertical sum** of individual demand (due to non-rivalry).



Public Goods and Common Pool Resource Goods

Common pool resource goods

- **Non-excludable** and **rival** in consumption
- Tragedy of the commons
 - Individuals act in self-interest $\Rightarrow$ Consume as much as possible (Non-excludability)
 - Resources are scarce
 - Resources are over-consumed $\Rightarrow$ Depletion

